

Comparing Precipitation Patterns in Seattle and London (2018–2022)

We often hear that Seattle is a very rainy city, but how rainy is it compared to other cities? To answer this question, I decided to compare Seattle's precipitation data with London's, another city known for its rainy weather. Using data from 2018 to 2022, I wanted to determine which city receives more precipitation and how frequently it occurs.

The data used in this project came from the National Oceanic and Atmospheric Administration (NOAA). NOAA provides weather data collected from stations around the world, which record different weather measurements, including temperature, precipitation, and snowfall.

For this project, I used daily precipitation measurements from weather stations in London, covering January 1, 2018, through December 31, 2022. For Seattle, my professor provided a GitHub link where I could access the dataset. The Seattle dataset originally contained additional weather measurements that were not relevant to my research question, so those columns were removed during the data-cleaning process.

To answer my research question, I first had to process and clean the data. During the initial analysis, I checked how many weather stations were represented in each dataset to better understand the data being compared. I also noticed that the date columns were not recognized as dates, so I converted them to a datetime format.

Since the datasets were separate, I combined them into one DataFrame to make it easier to visualize and compare the data. The combined dataset contained 3,652 entries, but the precipitation column had only 3,435 non-null values, indicating that some precipitation measurements were missing. To handle these missing values, I calculated the average precipitation for the same day of the year across the available years, separately for each city, and used those averages to fill in the missing values.

Additional columns, such as day of the year, month, and year, were created from the dates to make it easier to group and analyze the data by different time periods. This allowed me to compare precipitation patterns on a daily, monthly, and yearly basis for Seattle and London.

Once the data cleaning was completed and I had all the values needed for my analysis, I decided to follow two approaches. The first was to compare how frequently it rains in each city by counting the number of days with precipitation. The second was to compare how much precipitation each city received over the same period.

During the five years analyzed, or 1,826 days to be more precise, London experienced 634 days with precipitation, while Seattle had 999 days. This means that Seattle experienced precipitation on approximately 54.7% of the days, compared to only 34.7% in London. These results show that precipitation occurred much more frequently in Seattle than in London during this period.

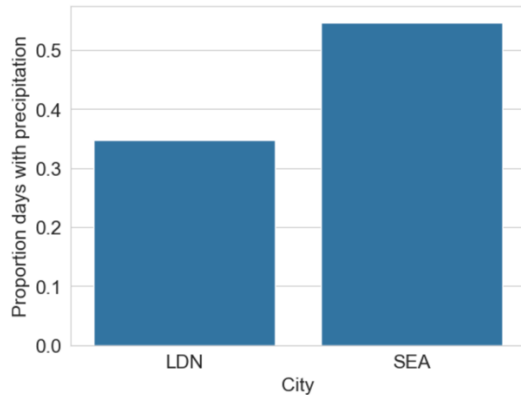


Figure 1. Proportion of days with precipitation in Seattle and London.

However, when comparing the average daily precipitation, both cities show very similar results. London had a daily average of 0.112 inches, while Seattle had 0.113 inches. Over the five-year period, London received approximately 205.62 inches of precipitation, compared to 206.83 inches in Seattle.

Looking at the monthly precipitation distribution, we can also observe that Seattle receives more precipitation during the winter months and very little during the summer. London, on the other hand, generally receives more precipitation than Seattle during the other seasons.

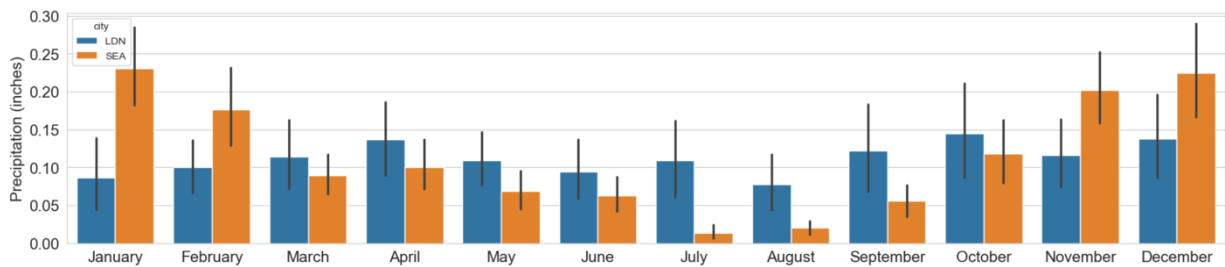


Figure 2. Average monthly precipitation in Seattle and London.

So, does it rain more in Seattle than in London? The answer depends on how we measure it. If we consider the number of days with precipitation, the answer is yes, Seattle is rainier than London. However, when comparing the total amount of precipitation, both cities show very similar results over the five-year period.

References:

National Oceanic and Atmospheric Administration. (n.d.). Climate Data Online (CDO). National Centers for Environmental Information. [ncei.noaa.gov](https://www.noaa.gov/data/monitoring-assessments/climate-data/)

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